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Summary

Computer Science student (FAST-NUCES) with hands-on full-stack development and AI/ML optimization expertise. Proficient in Python, C++, and JavaScript; specialized in constraint satisfaction algorithms, REST API design, and scalable web architecture. Proven track record in delivering 3+ deployed applications and 95

Education

FAST National University of Computer & Emerging Sciences

Islamabad, Pakistan

Bachelor of Science in Computer Science

Aug 2024 – Jun 2028

- **Relevant Coursework:** Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Operating Systems, Probability & Statistics

Lawrence College, Ghora Gali

Murree, Pakistan

Intermediate in Sciences

Aug 2022 – Jun 2024

Experience

AI & Machine Learning Engineer

Islamabad, Pakistan

Freelance

February 2025 – Present

- Engineered intelligent optimization solutions using **Python** and **Genetic Algorithms**, automating university scheduling across 200+ course sections with 95% constraint satisfaction
- Developed CSP-based scheduling solvers using **backtracking**, **arc consistency (AC-3)**, and heuristic-driven search, cutting computation time by **70%** over brute-force baselines
- Designed modular AI pipelines incorporating **A***, **Dijkstra's**, and graph-traversal algorithms, delivering reusable components for pathfinding and decision-making systems

Full Stack Web Developer

Islamabad, Pakistan

Freelance

February 2025 – Present

- Built and deployed **3+** client-facing web applications using **React.js**, **Next.js**, **Node.js**, and **Flask** with end-to-end RESTful API integration
- Optimized front-end rendering pipelines and implemented lazy-loading strategies, reducing average page-load time by **40%** across all deliverables
- Integrated third-party APIs for payment processing, analytics dashboards, and real-time data feeds, serving **500+** **concurrent users** with sub-second response times

Education & Mentoring

Islamabad, Pakistan

Junior Coderz

February 2025 – Present

- Mentored **50+** **students** in Python, web development, and computational thinking through structured, hands-on workshop curricula
- Facilitated interactive coding sessions on introductory AI concepts and logical problem-solving, boosting student engagement by **30%**
- Translated complex technical topics into accessible, activity-driven lessons tailored for beginner and intermediate learners

Projects

AI-Timetable-Optimizer | *Python, CSP, Backtracking, Arc Consistency* | 🐙

2025

- Built an intelligent **Constraint Satisfaction Problem** solver for automated timetable generation with backtracking and **AC-3** propagation
- Integrated **MRV** and **LCV** heuristics, pruning **70%** of the search space compared to naïve enumeration
- Designed a plug-in constraint engine supporting both hard and soft scheduling rules for real-world academic environments

- Engineered a multi-objective course-scheduling optimizer using **NSGA-II-inspired** selection with adaptive crossover and mutation operators
- Handled room-capacity, instructor-availability, and time-slot conflict constraints across **200+ sections**, achieving Pareto-optimal rankings
- Benchmarked fitness convergence over **500+ generations**, demonstrating consistent constraint satisfaction above **95%**

- Created an interactive GUI platform visualizing **4 search algorithms** on dynamic grid maps with real-time step-by-step animation
- Implemented configurable heuristic functions and weighted-graph support, enabling side-by-side performance benchmarking

- Architected a dual-column console editor using **doubly linked lists** for O(1) insertion/deletion with undo-redo and clipboard support
- Applied SOLID design principles with structured file I/O, robust memory management, and comprehensive error-handling mechanisms

Technical Skills

Languages: Python, C++, Java, C#, JavaScript, SQL, HTML/CSS, x86 Assembly

Frameworks & Libraries: React.js, Next.js, Flask, Node.js, NumPy, Pandas, Scikit-learn, Pygame, SFML

Developer Tools: Git, GitHub, VS Code, Visual Studio, Docker, Postman, L^AT_EX, Linux (Ubuntu)

Core Competencies: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, RESTful API Design, Optimization Algorithms, Constraint Satisfaction, Object-Oriented Design, Software Engineering, Agile/Scrum

Soft Skills: Problem-Solving, Communication, Collaboration, Analytical Thinking, Adaptability, Time Management, Attention to Detail